Difference Between Skewness and Kurtosis

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Understanding the shape of data is a key part of **Data Science**. It helps you identify where most of the information is concentrated and spot any unusual values (outliers). This guide explains the concepts of *skewness* and *kurtosis* —important tools for analyzing data.

You'll learn:

- What are skewness and kurtosis.
- How they describe the shape of data.
- Different types of skewness and kurtosis.
- How to interpret them.
- Their relationship and significance in statistics.

By the end, you'll understand these concepts clearly and be able to apply them to analyze data distributions effectively.

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What is Skewness?

Skewness is a statistical measure that assesses the asymmetry of a **probability distribution**. It quantifies the extent to which the data is skewed or shifted to one side.

Positive skewness indicates a longer tail on the right side of the distribution, while negative skewness indicates a longer tail on the left side. Skewness helps in understanding the shape

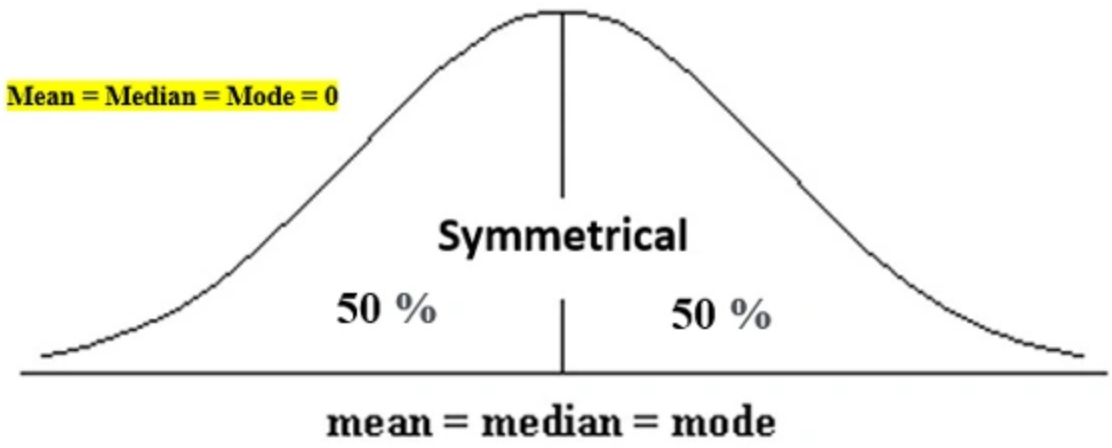
and outliers in a dataset.

Depending on the model, skewness in the values of a specific independent variable (feature) may violate model assumptions or diminish the interpretation of feature importance.

A probability distribution that deviates from the symmetrical normal distribution (bell curve) in a given set of data exhibits skewness, which is a measure of asymmetry in statistics.

A skewed data set, typical values fall between the first quartile (Q1) and the third quartile (Q3).

The normal distribution helps to know a skewness. When we talk about normal distribution, data symmetrically distributed. The symmetrical distribution has zero skewness as all measures of a central tendency lies in the middle.



Skewness and Kurtosis m=m=m,Skewness and Kurtosis

In a symmetrically distributed dataset, both the left-hand side and the right-hand side have an equal number of observations. (If the dataset has 90 values, then the left-hand side has 45 observations, and the right-hand side has 45 observations.). But, what if not symmetrical distributed? That data is called asymmetrical data, and that time skewness comes into the picture.

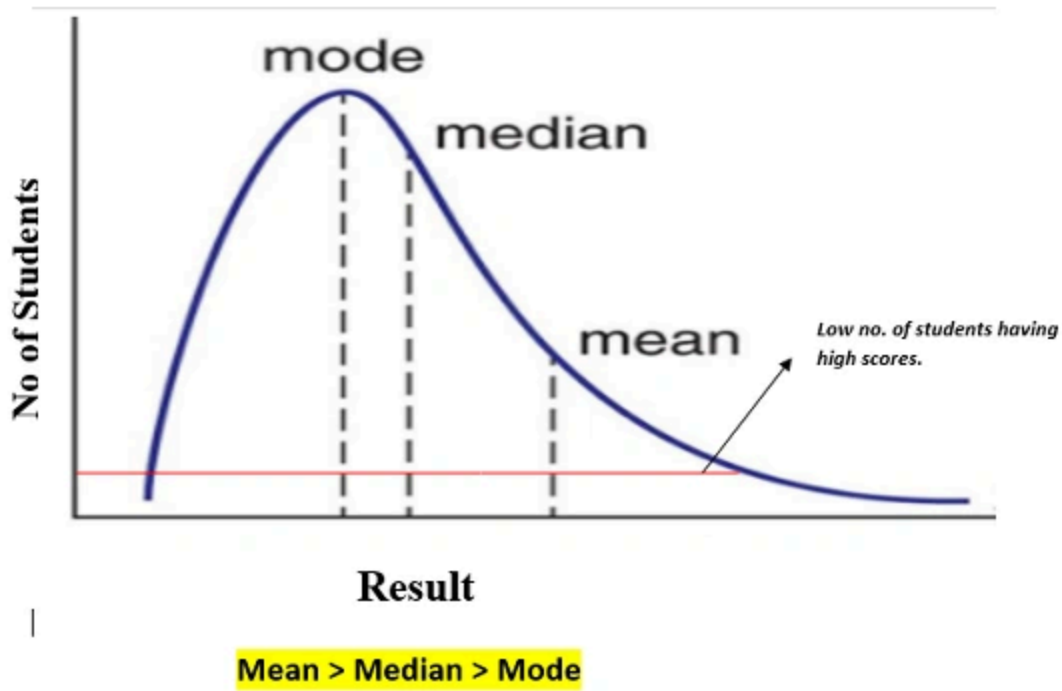
Also, Read about the [Skewness in Data and Its Impact on Data Analysis]

Types of Skewness

Positive Skewed or Right-Skewed (Positive Skewness)

In statistics, a positively skewed or right-skewed distribution has a long right tail. It is a sort of distribution where the measures are dispersing, unlike symmetrically distributed data where all measures of the central tendency (mean, median, and mode) equal each other. This makes

Positively Skewed Distribution a type of distribution where the mean, median, and mode of the distribution are positive rather than negative or zero.



1. Positive skewed or right-skewed

In positively skewed, the mean of the data is greater than the median (a large number of data-pushed on the right-hand side). In other words, the results are bent towards the lower side. The mean will be more than the median as the median is the middle value and mode is always the most frequent value.

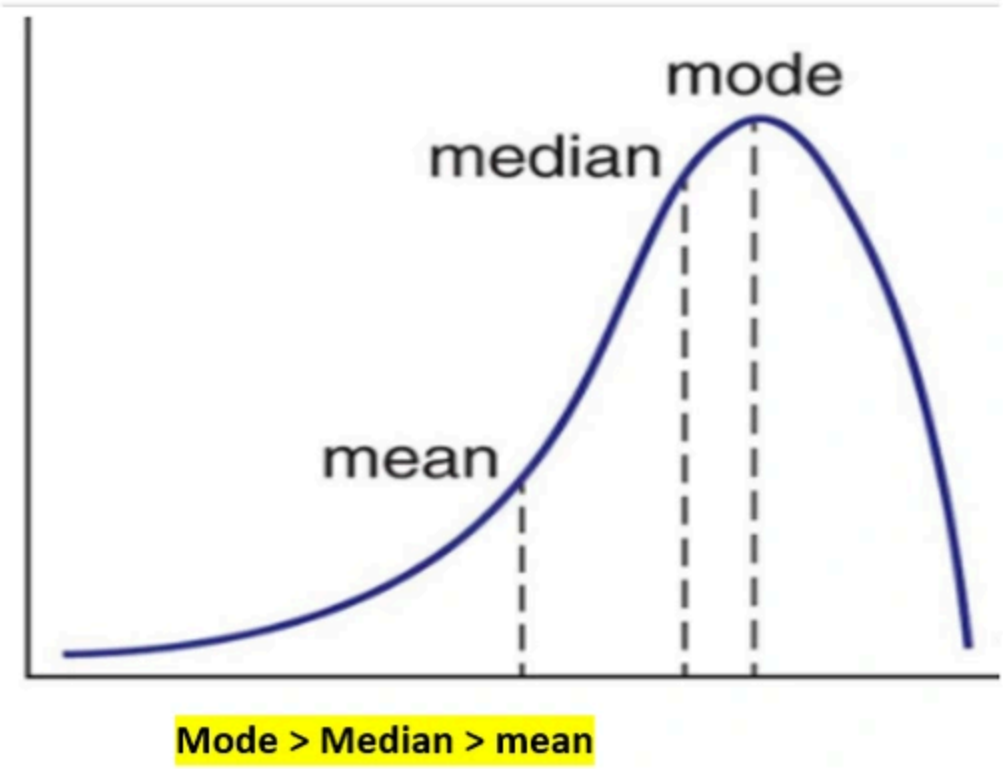
Extreme positive skewness is not desirable for a distribution, as a high level of skewness can cause misleading results. The data transformation tools are helping to make the skewed data closer to a normal distribution. For positively skewed distributions, the famous transformation is the log transformation. The log transformation proposes the calculations of the natural logarithm for each value in the dataset.

Read More about the Feature Transformations in Data Science

Negative Skewed or Left-Skewed (Negative Skewness)

A distribution with a long left tail, known as negatively skewed or left-skewed, stands in complete contrast to a positively skewed distribution. skewness and kurtosis in statistics, negatively skewed distribution refers to the distribution model where more values are plots on the right side of the graph, and the tail of the distribution is spreading on the left side.

In negatively skewed, the mean of the data is less than the median (a large number of data-pushed on the left-hand side). Negatively Skewed Distribution is a type of distribution where the mean, median, and mode of the distribution are negative rather than positive or zero.



Negative skewed or left-skewed

Median is the middle value, and mode is the most frequent value. Due to an unbalanced distribution, the median will be higher than the mean.

How to Calculate the Skewness Coefficient?

Various methods can calculate **skewness**, with Pearson's coefficient being the most commonly used method.

Pearson's first coefficient of skewness

To calculate skewness values, subtract the mode from the mean, and then divide the difference by standard deviation.

$$\text{Pearson's first coefficient} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

Pearson's first coefficient of skewness

As Pearson's correlation coefficient differs from -1 (perfect negative linear relationship) to +1 (perfect positive linear relationship), including a value of 0 indicating no linear relationship, When we divide the covariance values by the standard deviation, it truly scales the value down to a limited range of **-1 to +1.** That accurately shows the range of the correlation values.

Pearson's first coefficient of skewness is helping if the data present high mode. However, if the data exhibits low mode or multiple modes, it is preferable not to use Pearson's first coefficient, and instead, Pearson's second coefficient may be superior, as it does not depend on the mode.

Pearson's second coefficient of skewness
subtract the median from the *mean*, *multiply the difference by 3, and divide the product by the standard deviation.*

$$\text{Pearson's second coefficient} = \frac{3\,(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

$$\text{Mean} - \text{Mode} \approx 3\,(\text{Mean} - \text{Median})$$

Pearson's first coefficient of skewness

Rule of thumb:

- **For skewness values between -0.5 and 0.5, the data exhibit approximate symmetry.**
- **Skewness values within the range of -1 and -0.5 (negative skewed) or 0.5 and 1(positive skewed) indicate slightly skewed data distributions.**
- **Data with skewness values less than -1 (negative skewed) or greater than 1 (positive skewed) are considered highly skewed.**

What is Kurtosis?

Kurtosis is a statistical measure that quantifies the shape of a probability distribution. It provides information about the tails and peakedness of the distribution compared to a normal distribution.

Positive kurtosis indicates heavier tails and a more peaked distribution, while negative kurtosis suggests lighter tails and a flatter distribution. Kurtosis helps in analyzing the characteristics and outliers of a dataset.

The measure of Kurtosis refers to the tailedness of a distribution. Tailedness refers to how often the outliers occur.

Peakedness in a data distribution is the degree to which data values are concentrated around the mean. Datasets with high kurtosis tend to have a distinct peak near the mean, decline rapidly, and have heavy tails. Datasets with low kurtosis tend to have a flat top near the mean rather than a sharp peak.

In finance, kurtosis is used as a measure of financial risk. A large kurtosis is associated with a high level of risk for an investment because it indicates that there are high probabilities of extremely large and extremely small returns. On the other hand, a small kurtosis signals a moderate level of risk because the probabilities of extreme returns are relatively low.

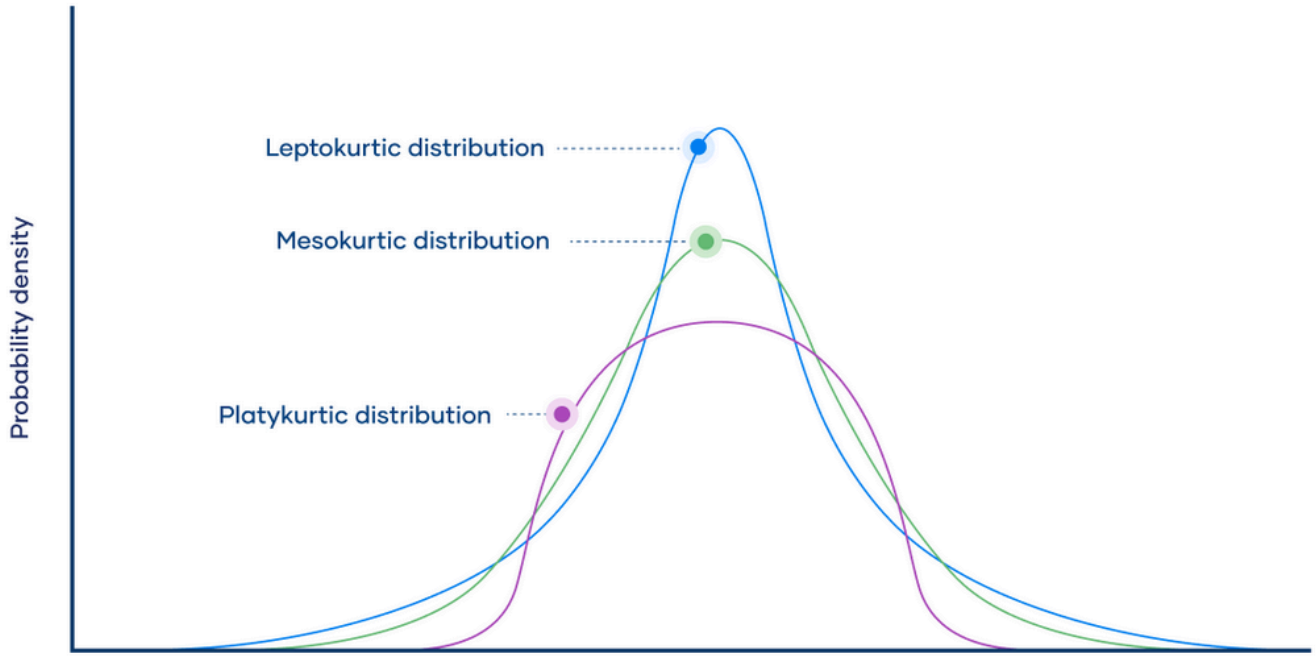
What is Excess Kurtosis?

In statistics and probability theory, researchers use excess kurtosis to compare the kurtosis coefficient with that of a normal distribution. Excess kurtosis can be positive (Leptokurtic distribution), negative (Platykurtic distribution), or near zero (Mesokurtic distribution). Since normal distributions have a kurtosis of 3, excess kurtosis is calculated by subtracting kurtosis by 3.

Excess kurtosis = Kurt – 3

Types of Kurtosis

Kurtosis is a statistical measure that describes the shape of a probability distribution's tails relative to its peak. There are three main types of kurtosis:



Skewness and Kurtosis

1. **Mesokurtic**: A distribution with mesokurtic kurtosis has a similar peak and tail shape as the normal distribution. It has a kurtosis value of around 0, indicating that its tails are neither too heavy nor too light compared to a normal distribution.
2. **Leptokurtic**: A distribution with leptokurtic kurtosis has heavier tails and a sharper peak than the normal distribution. It has a positive kurtosis value, indicating that it has more extreme outliers than a normal distribution. This type of distribution is often associated with higher peakedness and a greater probability of extreme values.
3. **Platykurtic**: A distribution with platykurtic kurtosis has lighter tails and a flatter peak than the normal distribution. It has a negative kurtosis value, indicating that it has fewer extreme outliers than a normal distribution. This type of distribution is often associated with less peakedness and a lower probability of extreme values.

Checkout this article about the [Importance of Skewness, Kurtosis, Co-efficient of Variation]

Skewness and Kurtosis Formula

Skewness and kurtosis are two statistical measures that describe the shape of a distribution. Let's look at skewness and kurtosis formula in the next section!

Skewness Formula

Skewness measures the asymmetry of a distribution. A symmetrical distribution has a skewness of zero. Positive skewness indicates that the right tail of the distribution is longer or fatter than the left tail, while negative skewness indicates the opposite.

The formula for skewness (often denoted by $\gamma 1$ γ 1) for a sample is:

γ 1=(n −1)(n −2) n ∑ i =1 n (sxi − x ¯)3

Where:

- n n is the number of observations in the sample
- xi x i is the ith observation
- $x¯$ x ¯ is the sample mean
- s s is the sample standard deviation

Kurtosis Formula

Kurtosis measures the peakedness or flatness of a distribution relative to the normal distribution. A normal distribution has a kurtosis of 3, known as the excess kurtosis. Deviations from this value indicate how much the distribution deviates from the normal, with positive excess kurtosis indicating a more peaked distribution and negative excess kurtosis indicating a flatter one.

The formula for kurtosis (often denoted by $\gamma 2$ γ 2) for a sample is:

γ 2=(n −1)(n −2)(n −3) n (n +1)∑ i =1 n (sxi − x ¯)4−(n −2)(n −3)3(n −1)2

Where:

- n n is the number of observations in the sample
- xi x i is the ith observation
- x¯ x ¯ is the sample mean
- s s is the sample standard deviation

These formulas give the sample skewness and kurtosis. For population skewness and kurtosis, the divisor n n in the formulas is replaced with n−1 n −1 and n−2 n −2, respectively.

Difference Between Skewness and Kurtosis

Skewness	Kurtosis
Skewness measures the asymmetry of a probability distribution	Kurtosis measures the tailedness or peakedness of a probability distribution
Positive skew indicates a right-skewed distribution, with the tail extending to the right	Positive kurtosis indicates a distribution with heavier tails, often referred to as “leptokurtic”
Negative skew indicates a left-skewed distribution, with the tail extending to the left	Negative kurtosis indicates a distribution with lighter tails, often referred to as “platykurtic”
A skewness value of zero indicates a symmetric distribution	A kurtosis value of zero indicates a distribution similar to the normal distribution, often referred to as “mesokurtic”
Used to identify the direction and degree of asymmetry	Used to identify the presence of outliers or extreme values
Sensitive to changes in the tails of the distribution	Sensitive to changes in the center and shoulders of the distribution
Commonly used in fields such as economics, finance, and social	Commonly used in statistics, engineering, and physical sciences

Skewness	Kurtosis
sciences	
Examples: income distribution, stock returns	Examples: particle physics, image processing

Conclusion

Skewness and kurtosis are important tools for analyzing the shape of data distributions, giving us a clearer picture of how data behaves.

- **Skewness** tells us if the data leans to one side:
  - Positive skew means the tail is longer on the right.
  - Negative skew means the tail is longer on the left.
  - Skewness between -0.5 and 0.5 indicates a nearly symmetrical distribution.
- **Kurtosis** describes the “tails” and “peaks” of the distribution:
  - High kurtosis (leptokurtic) means heavy tails with more outliers.
  - Low kurtosis (platykurtic) means light tails with fewer outliers.
  - Near-zero excess kurtosis (mesokurtic) is close to a normal distribution.

Together, these measures help identify patterns, outliers, and whether the data is suitable for statistical models. Skewed data, for example, might require transformation to resemble a normal distribution for better model performance, especially in regression.

By using skewness and kurtosis, you can better understand your data and make informed decisions in your analysis.

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Responses From Readers

Hi SUVARNA3, I am a data scientist at KNIME and I really like your article! How can I get in touch with you?

yes I am intersting for you lesson today.

thanks, keep it up

Frequently Asked Questions

A. Skewness measures the symmetry of a data distribution, indicating if it leans left or right. Kurtosis evaluates the “tailedness” of the distribution, showing if data has heavy or light tails compared to a normal distribution.

A. Kurtosis is a statistical measure that describes the shape of a data distribution’s tails and peak. It helps determine whether the distribution has heavy tails (more outliers) or light tails (fewer outliers).

A. Skewness near zero indicates symmetry; positive values mean a right-skewed distribution, and negative values indicate left skew. For kurtosis, high values signal heavy tails, while low values indicate light tails.

A. Skewness is a measure of the asymmetry of a data distribution. It reveals if data values are concentrated more on one side, causing the distribution to tilt left (negative skew) or right (positive skew).

A. Skewness measures how lopsided your data is. Kurtosis measures how much data is in the tails (outliers).

Ideally: Skewness is close to 0, and kurtosis is close to 3.

Usually OK: Skewness and kurtosis are between -2 and +2.

A. The three types are:

– **Positive Skew:** Tail extends to the right.

– **Negative Skew:** Tail extends to the left.

– **Symmetrical (No Skew):** Data is evenly distributed.